

The Euler-Lagrange Equations: An Action-Packed Tango to Unravel the Elegance of Lagrangian Mechanics

Chris Wang

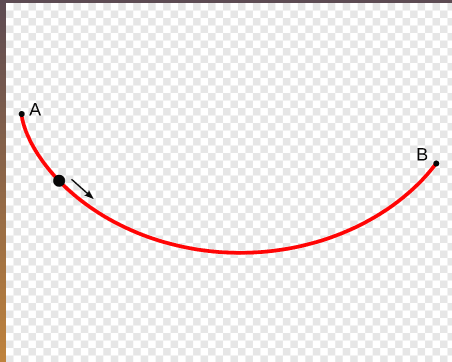
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Why Do We Care About Lagrangian Mechanics?

- ▶ Two pictures of mechanics
 - ▶ Newton — forces
 - ▶ Lagrange/Hamilton — energy, conservation laws

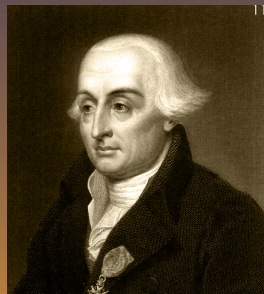
A Brief History

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 - ▶ Lagrange: 1755 → Euler
 - ▶ Euler + Maupertuis: Action
- Lagrangian Mechanics, and the *Mécanique analytique*



Action, and Extremizing It

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L is the *Lagrangian*, $L = T - U$.

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- ▶ Hamilton's principle:

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- ▶ What conditions does L satisfy when $\delta S = 0$?

Deriving Conditions on L : Perturbations

- ▶ Perturbative approach in t :

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- ▶ Boundary conditions on ε : $dS/d\varepsilon|_{\varepsilon=0} = 0$.

Chain Rule: helo

► Since $S = S(q, \dot{q})$:

$$dS = \frac{\partial S}{\partial q} dq + \frac{\partial S}{\partial \dot{q}} d\dot{q} \quad \Longrightarrow \quad \frac{dS}{d\varepsilon} = \frac{\partial S}{\partial q} \frac{dq}{d\varepsilon} + \frac{\partial S}{\partial \dot{q}} \frac{d\dot{q}}{d\varepsilon}. \quad (4)$$

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- ▶ Evaluate partial derivatives:

$$\frac{dS}{d\varepsilon} = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} \right) dt. \quad (5)$$

From Parts, the Whole (Equation)

- Integration by parts yields

$$\frac{dS}{d\varepsilon} = \int_{t_1}^{t_2} \eta \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) dt. \quad (6)$$

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- But! $dS/d\varepsilon|_{\varepsilon=0} = 0$, so

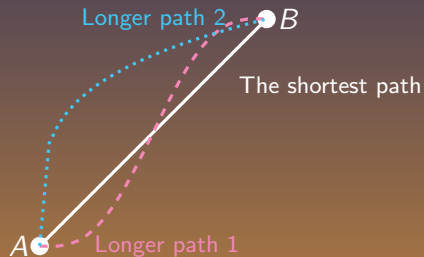
$$0 = \frac{dS}{d\varepsilon} \Big|_{\varepsilon=0} = \int_{t_1}^{t_2} \eta \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) dt. \quad (7)$$

- η is arbitrary!

The Final Form

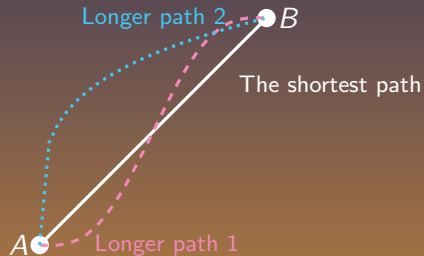
$$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0.$$

Let's Talk Geodesics: A Nifty Application (You Do It!)

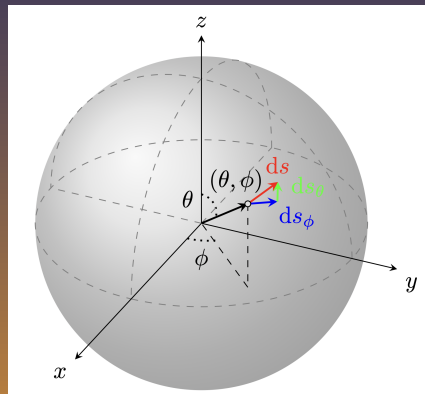


A geodesic in Euclidean space is a straight line.

Let's Talk Geodesics: A Nifty Application (You Do It!)



A geodesic in Euclidean space is a straight line.



A geodesic in spherical space is a great circle.

Retrieving Familiar Force Laws (You Do It!)

→ Gravity:

$$\mathcal{L} = \frac{1}{2}m\dot{r}^2 + \frac{GMm}{r^2},$$

$$\implies m\ddot{r} = -\frac{GMm}{r^2}.$$



m

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→ Pendulum:

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 + mg\ell \cos \theta,$$

$$\implies \ddot{\theta} \simeq -\frac{g}{\ell}\theta.$$

